AquaPoint: Self-Supervised Cross-Domain Framework for Robust Underwater Local Features

Boyong Zhao[1], Yuan Shen[1], Pui Yi Chui[2], Yuzhe Li[3], Junfeng Wu[1,†]

***Abstract*— Simultaneous Localization and Mapping (SLAM) is essential for autonomous underwater vehicles (AUVs). However, severe hydro-optical scattering and non-linear geometric distortions frequently cause feature tracking failures in existing methods. To address this, we introduce AquaPoint, a self-supervised cross-domain framework tailored for robust underwater local feature extraction. Our framework comprises two cascaded modules. First, the Physics-Guided Domain Bridging (PGDB) module explicitly mitigates both photometric and structural domain gaps by synergizing physics-consistent hydro-optical degradation synthesis with OceanSim simulations. Second, to overcome ground truth scarcity and adapt to real-world hydro-optical dynamics, the Geometry-Consistent Descriptor Alignment (GCDA) module leverages air-calibrated intrinsics and multi-view geometric verification to mine verified pseudo-correspondences for contrastive learning. Evaluations on AQUALOC and FLSea demonstrate that AquaPoint consistently surpasses existing baselines, yielding superior absolute numbers of matches and inliers with denser spatial distributions. Furthermore, integrating AquaPoint into a decoupled pure-odometry ORB-SLAM3 configuration validates its trajectory tracking robustness, achieving 100% trajectory completion ratio across Structure sequences and reducing the Absolute Trajectory Error (ATE) by up to 75.4% compared to SuperPoint under pure-vision conditions.**

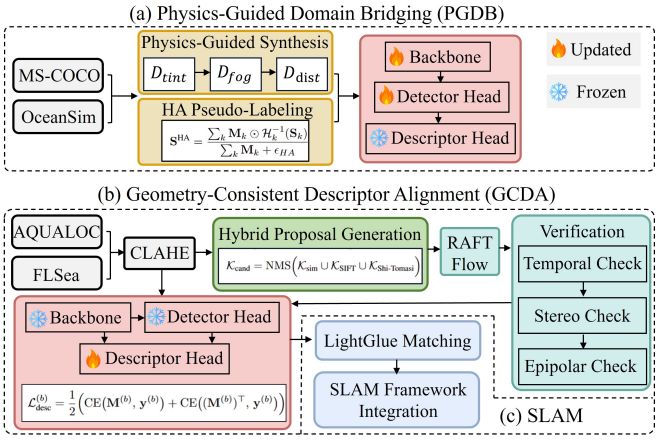


Fig. 1: Overall architecture of AquaPoint framework, which consists of the PGDB module and the GCDA module. PGDB bridges the photometric and structural gaps, while GCDA addresses the scarcity of underwater ground truth and the descriptor mismatch under real hydro-optical dynamics. The learned features are integrated into ORB-SLAM3 with LightGlue matching for downstream evaluation.

I. Introduction

Autonomous Underwater Vehicles (AUVs) serve as pivotal platforms for deep-sea resource exploration and marine infrastructure inspection, fundamentally depending on Simultaneous Localization and Mapping (SLAM) for autonomous navigation in Global Navigation Satellite System (GNSS) denied regions [1]–[3]. In Visual-Inertial SLAM systems, the front-end module, which is responsible for feature extraction, governs the observability and convergence of back-end state estimation [4]–[6].

Nevertheless, the marine medium introduces optical and geometric challenges. Wavelength-dependent absorption and scattering induce severe contrast attenuation, whereas suspended particles generate time-varying noise [7]. Additionally, interfacial refraction at waterproof housings violates the conventional pinhole assumption, yielding non-linear geometric distortions [8]. These degradation factors diminish feature repeatability, frequently leading to tracking failures and cumulative drift.

Current feature extraction paradigms generally bifurcate into two categories, both exhibiting inherent vulnerabilities in marine deployments. Classical hand-crafted features, such as Scale-Invariant Feature Transform (SIFT) [9] and Oriented FAST and Rotated BRIEF (ORB) [10], rely heavily on local gradients, rendering them highly susceptible to turbidity and illumination fluctuations [1], [6]. Conversely, learning-based models like SuperPoint [11] excel on terrestrial benchmarks but experience substantial efficacy drops underwater due to profound photometric and structural domain discrepancies. Although data synthesis attempts to bridge this gap, the accurate emulation of complex hydro-optical dynamics remains intrinsically challenging [12], [13]. Crucially, terrestrial-derived synthetic imagery lacks the topological priors characteristic of authentic seabed terrains, potentially misguiding networks toward erroneous spatial representations [14].

To resolve these limitations, we introduce AquaPoint, a self-supervised cross-domain framework tailored for underwater local features. Built upon the lightweight SuperPoint backbone, AquaPoint focuses on improving the underwater robustness of an efficient sparse feature baseline, rather than introducing a heavier transformer architecture [15]. By coupling physics-informed rendering with rigorous geometric constraints, the proposed framework fortifies both detector repeatability and descriptor resilience. The framework comprises two cascaded modules. First,

[1]School of Data Science, The Chinese University of Hong Kong, Shenzhen, China.
[2]School of Life Sciences, The Chinese University of Hong Kong, China.
[3]State Key Laboratory of Synthetical Automation for Process Industries, Northeastern University, Shenyang, China.
[†]Corresponding author: junfengwu@cuhk.edu.cn.

the Physics-Guided Domain Bridging (PGDB) module establishes a physics-consistent degradation synthesis combined with photogrammetry-reconstructed simulations to bridge the photometric and structural gaps. Subsequently, the Geometry-Consistent Descriptor Alignment (GCDA) module leverages air-calibrated intrinsics alongside multi-view geometric verification to filter outliers, thereby mining verified pseudo-correspondences for contrastive learning.

To quantify the practical impact on state estimation, we integrate AquaPoint into ORB-SLAM3 [4] and evaluate it under a decoupled pure-odometry configuration. This design intentionally isolates the influence of the front-end features and serves as a downstream evaluation framework to validate feature robustness.

In summary, the main contributions are as follows:

- We propose the Physics-Guided Domain Bridging (PGDB) module to explicitly bridge the terrestrial-to-underwater photometric and structural gaps. By coupling physics-consistent hydro-optical degradation synthesis with photogrammetry-reconstructed simulation fine-tuning, PGDB enables robust, annotation-free detector transfer.
- We develop AquaPoint, an annotation-free, fully self-supervised cross-domain local feature framework for underwater matching. Extensive experiments show that AquaPoint boosts verified inliers by 76.5% on AQUALOC and by 488.9% on FLSea compared to the SuperPoint baseline.
- We integrate AquaPoint into the ORB-SLAM3 front-end for downstream evaluation. Under a decoupled pure-odometry configuration, AquaPoint substantially fortifies tracking resilience against severe visual degradation, reducing ATE by 30.8% on average and boosting the trajectory completion ratio to 100% on challenging sequences, achieving state-of-the-art performance on the Tank Structure sequences. Code and datasets will be made publicly available.

II. Related Work

Underwater image enhancement is commonly used to mitigate absorption and scattering before downstream perception [16]. Physics-based methods such as Sea-Thru [7] recover scene appearance by estimating optical parameters, while data-driven methods such as WaterGAN [12] improve visual quality through image translation. However, improved perceptual quality does not necessarily translate into reliable SLAM performance, since dense depth requirements, hallucinated textures, or temporal inconsistency may undermine stable matching and tracking [16]–[18].

To bypass the constraints of front-end restoration, recent underwater SLAM studies have sought to improve robustness through system-level compensation [2], [6], [19]. Conventional operators (e.g., SIFT [9], ORB [10]) demonstrate critical reliability drops under turbid and dynamically illuminated conditions [19], while learning-based counterparts [11], [15] confront severe topological domain shifts in unstructured benthic settings. To mitigate this feature fragility, system-level compensation strategies have emerged. These include tightly-coupled visual-inertial fusion (e.g., SVIn2 [6]) to bound drift during sparse visual tracking, and the incorporation of refractive ray-tracing models into bundle adjustment to rectify non-linear distortions [8]. Although empirical evaluations indicate improved precision, these back-end interventions invariably impose substantial computational overhead and mandate rigorous calibration procedures [8]. More importantly, they do not directly resolve the intrinsic fragility of front-end local features under severe hydro-optical degradation.

Motivated by these limitations, we focus on learning robust local features directly for the SLAM front-end. Unlike existing methods that either rely on unstable preprocessing or address degradation only at the system level, AquaPoint introduces a self-supervised cross-domain framework that does not depend on scarce underwater ground truth. By combining physics-guided synthesis with multi-view geometric verification, AquaPoint explicitly bridges the photometric and structural domain gaps, improving feature resilience without adding overhead to the SLAM back-end.

III. Methodology

A. Overview

We propose a framework for learning robust local features in underwater degraded environments, explicitly addressing three primary challenges: photometric domain gaps, structural discrepancies, and ground truth scarcity. We build our framework upon SuperPoint [11] as the base detector–descriptor network, and denote it as $f_{\theta_{\text{base}}}(\cdot)$. As illustrated in Fig. 1, our framework comprises two cascaded modules. First, the PGDB module explicitly mitigates both photometric and structural discrepancies by synergizing physics-consistent degradation synthesis with photogrammetry-reconstructed OceanSim simulations. Subsequently, to directly address ground truth scarcity and adapt to real-world hydro-optical dynamics, the GCDA module leverages air-calibrated intrinsics and multi-view geometric verification. By mining verified pseudo-correspondences from unlabeled underwater footage, it supervises a contrastive alignment of descriptors.

B. Physics-Guided Domain Bridging

1) Physics-Consistent Appearance Synthesis: To harness rich texture priors from Microsoft Common Objects in Context (MS-COCO) [20] while narrowing photometric domain gaps, we employ a physics-based synthesis pipeline to transform source images $\mathbf{I}_{\text{src}} \in [0,1]^{3\times H\times W}$ into the underwater domain. The formation process is modeled as a composition of differentiable degradation operators:

$$\mathbf{I}_{\text{uw}} = \mathcal{D}_{\text{dist}}(\mathcal{D}_{\text{fog}}(\mathcal{D}_{\text{tint}}(\mathbf{I}_{\text{src}}))), \quad \mathbf{I}_{\text{uw}} \in [0,1]^{3\times H\times W} \tag{1}$$

where $\mathcal{D}_{\text{tint}}$, $\mathcal{D}_{\text{fog}}$, and $\mathcal{D}_{\text{dist}}$ correspond to color absorption, scattering fog, and geometric distortion, respectively.

Color absorption ($\mathcal{D}_{\text{tint}}$): Water selectively absorbs light, with red attenuating faster than blue and green. For a pixel

coordinate $\mathbf{p}$, we define this operator via a channel-wise scaling vector $\boldsymbol{\tau} = [\tau_R, \tau_G, \tau_B]$. The detailed parameter settings are provided in Sec. IV:

$$\mathcal{D}_{\text{tint}}(\mathbf{I}_{\text{src}}, \mathbf{p}) = \mathbf{I}_{\text{src}}(\mathbf{p}) \odot \tau \tag{2}$$

Scattering fog ($\mathcal{D}_{\text{fog}}$): Since MS-COCO lacks depth information, we generate a vertical gradient pseudo-depth $d(\mathbf{p})$ to approximate line-of-sight variations:

$$d(\mathbf{p}) = d_{\max} - (d_{\max} - d_{\min})\frac{y}{H}, \tag{3}$$

where $d_{\max}$ and $d_{\min}$ represent the maximum and minimum pseudo-depth, y denotes the vertical coordinate of the pixel $\mathbf{p}$. Following the Jaffe-McGlamery model [21], the fog effect is synthesized by blending the signal with ambient light $\mathbf{B}$ based on transmission $t(\mathbf{p}) = e^{-\beta d(\mathbf{p})}$:

$$\mathcal{D}_{\text{fog}}(\mathbf{I}_{\text{src}}, \mathbf{p}) = \mathbf{I}_{\text{src}}(\mathbf{p}) \cdot t(\mathbf{p}) + \mathbf{B} \cdot (1 - t(\mathbf{p})), \tag{4}$$

where β represents the attenuation coefficient and $\mathbf{B} = [B_R, B_G, B_B]$ denotes the ambient backscatter light.

Geometric distortion ($\mathcal{D}_{\text{dist}}$): To accurately simulate nonlinear optical aberrations, we employ a standard first-order radial distortion model. Briefly, a target pixel coordinate $\mathbf{p}$ is unprojected to the camera's normalized plane using the intrinsic matrix, radially distorted by a coefficient k_1, and reprojected to derive the corresponding source sampling coordinate $\mathbf{p}'$. The geometric distortion operator $\mathcal{D}_{\text{dist}}$ is subsequently formulated via spatial backward mapping, generating the distorted image by sampling the input image $\mathbf{I}_{\text{src}}$ at $\mathbf{p}'$:

$$\mathcal{D}_{\text{dist}}(\mathbf{I}_{\text{src}}, \mathbf{p}) = \mathcal{S}(\mathbf{I}_{\text{src}}, \mathbf{p}'), \tag{5}$$

where $\mathcal{S}(\cdot)$ denotes the interpolation sampling function.

We apply image-space, physics-guided underwater synthesis on MS-COCO rather than simulator-based rendering, since scaling scene-complete 3D asset construction and sensor-level rendering to COCO scale is prohibitively labor-intensive. In practice, the resulting COCO-Water data provide sufficiently realistic photometric cues for pretraining.

2) Homography-Based Pseudo-Labeling: To reduce reliance on manual annotations, we use Homography Adaptation (HA) [11] to aggregate detection responses. For each synthetic image, we sample a set of random homographies $\{\mathcal{H}_k(\cdot)\}_{k=1}^{N}$. For the input image $\mathbf{I}_{\text{uw}}$, we compute the warped image $\mathbf{I}_k = \mathcal{H}_k(\mathbf{I}_{\text{uw}})$ and its corresponding detector heatmap $\mathbf{S}_k = f_{\theta_{\text{base}}}(\mathbf{I}_k)$.

To prevent the network from learning artifacts at the boundaries of warped images, we define a pixel-level validity mask $\mathbf{M}_k = \mathcal{E}_r\big(\mathcal{H}_k^{-1}(\mathbf{1})\big) \in \{0,1\}^{H\times W}$, where $\mathcal{E}_r(\cdot)$ denotes binary morphological erosion [22] with a radius structuring element r, and $\mathbf{1} \in \{1\}^{H\times W}$ denotes an all-one mask. $\mathcal{H}^{-1}(\cdot)$ represents the inverse transform of homography $\mathcal{H}(\cdot)$. The heatmaps are unwrapped to align the original image and aggregated to form the final pseudo-label $\mathbf{S}^{\text{HA}}$:

$$\mathbf{S}^{\text{HA}} = \frac{\sum_k \mathbf{M}_k \odot \mathcal{H}_k^{-1}(\mathbf{S}_k)}{\sum_k \mathbf{M}_k + \epsilon_{HA}}, \tag{6}$$

where ϵ_{HA} is a small constant to ensure numerical stability. Keypoints $\mathcal{K}$ are then extracted from $\mathbf{S}^{\text{HA}}$ via Non-Maximum Suppression (NMS).

3) Training Objective: Following SuperPoint [11], the detector is optimized using a cross-entropy cell-classification objective. To suppress invalid regions caused by geometric warping, we compute a pixel-domain validity mask based on the warp support and downsample it to the detector grid. The final training objective $\mathcal{L}_{\text{total}}$ is the sum of the standard classification loss on the original image and the masked classification loss on the warped view.

4) Photogrammetry-Reconstructed Simulation Fine Tuning: While the physics-based synthesis pipeline addresses photometric domain gaps, terrestrial datasets intrinsically lack the complex benthic topography of authentic seabed terrains, leaving severe structural discrepancies unresolved. To explicitly mitigate this terrestrial-to-underwater structural domain gap, we fine-tune on OceanSim [14], which offers photogrammetry-reconstructed seafloor scenes and configurable sensor rendering with the same objective $\mathcal{L}_{\text{total}}$.

C. Geometry-Consistent Descriptor Alignment

To directly address ground truth scarcity and adapt to complex real-world hydro-optical dynamics, this module focuses on aligning the descriptor representation with unlabeled authentic underwater footage. We freeze the backbone and detector parameters to prevent distribution drift and preserve the structural priors learned in the PGDB module. Subsequently, we leverage air-calibrated intrinsics and multi-view geometric verification to mine verified pseudo-correspondences, constructing a self-supervised signal for contrastive learning.

1) Hybrid Proposal Generation: To ensure high feature recall across varying turbidity levels, we first apply Contrast Limited Adaptive Histogram Equalization (CLAHE) [23]. We then construct a hybrid candidate keypoints set $\mathcal{K}_{\text{cand}}$ by aggregating predictions from the simulation-adapted detector with classical operators. This strategy mitigates potential domain overfitting and ensures coverage in low-texture regions:

$$\mathcal{K}_{\text{cand}} = \text{NMS}\Big(\mathcal{K}_{\text{sim}} \cup \mathcal{K}_{\text{SIFT}} \cup \mathcal{K}_{\text{Shi-Tomasi}}\Big), \tag{7}$$

where $\mathcal{K}_{\text{sim}}$ denotes keypoints from the simulation fine-tuning network $f_{\theta_{sim}}(\cdot)$, supplemented by SIFT [9] keypoints $\mathcal{K}_{\text{SIFT}}$ and Shi-Tomasi [24] keypoints $\mathcal{K}_{\text{Shi-Tomasi}}$.

2) Multi-View Geometric Verification for Pseudo-Correspondence Mining: HA assumes planar scenes and is fragile under underwater scattering and temporal artifacts. We instead employ RAFT [25] to estimate dense forward and backward optical flows. RAFT's correlation volumes provide broader contextual awareness and naturally accommodate the complex 3D structures of benthic environments, yielding more robust temporal pseudo-correspondences in visually degraded marine conditions.

As illustrated in Fig. 2, a temporal match is accepted only if it satisfies the temporal forward-backward cycle consistency check. We use a strict pixel threshold τ_t to tolerate the

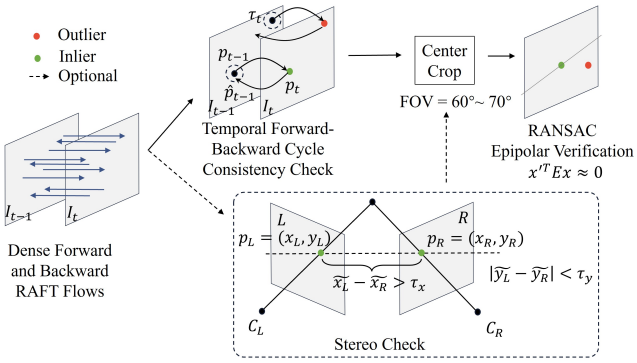


Fig. 2: Multi-View Geometric Verification

small localization deviations induced by underwater scattering and blur, while rejecting unstable correspondences [3]. Following the small-pixel gating strategy commonly used in underwater optical-flow-based VO [26], we set a stricter threshold of $\tau_t = 1.0$ px to tolerate such perturbations while favoring high-purity pseudo-correspondences.

Consistent with common practice in underwater SLAM systems [2], [6], we perform correspondence validation under the conventional single-viewpoint pinhole model using air-calibrated intrinsics and distortion parameters, treating it as a tractable approximation for epipolar verification. As discussed in [8], modeling the interfacial refraction at waterproof housings with depth-dependent refractive rays complicates calibration and incurs non-trivial computational overhead, often requiring dense depth priors. Accordingly, we retain the in-air undistortion and epipolar verification pipeline for correspondence mining. Since residual nonlinear warping is most pronounced toward the image periphery, where feature localization and matching stability deteriorate, we restrict the effective field of view (FOV) to approximately 60° to 70° by cropping peripheral regions, and conduct subsequent matching and epipolar verification within the remaining central area. This design keeps the correspondence mining procedure tractable and aligns with the main focus of AquaPoint, namely improving underwater feature robustness under photometric and structural domain gaps.

For datasets captured by stereo cameras with a horizontal baseline, we additionally impose constraints on the rectified coordinates $(\tilde{x}, \tilde{y})$. A match is verified if it remains approximately row-aligned and exhibits sufficient positive disparity in the rectified domain, i.e., $|\tilde{y}_L - \tilde{y}_R| < \tau_y$ and $\tilde{x}_L - \tilde{x}_R > \tau_x$, where subscripts L and R denote the left and right views. Since refraction leaves small residual row misalignment after rectification and near-zero disparity yields poorly conditioned depth estimates [8], we adopt a small vertical tolerance and a minimal positive-disparity gate, setting $\tau_y = 1.5$ px and $\tau_x = 1$ px in all stereo experiments.

3) InfoNCE-Based Contrastive Descriptor Alignment: Despite strict multi-view geometric verification, residual ambiguity may remain under scattering, blur, and repetitive textures. We therefore optimize the descriptor head with an InfoNCE [27] objective, which encourages verified positives to be close while using in-batch correspondences as negatives.

Let an image pair $(\mathbf{I}^0, \mathbf{I}^1)$ denote either temporal neighbors or rectified stereo views after multi-view geometric verification, indexed as view 0 and view 1. We denote the dense descriptor maps as $\mathbf{F}_b^0$ and $\mathbf{F}_b^1 \in \mathbb{R}^{C \times H_c \times W_c}$ for the b-th batch element, where C is the descriptor channel dimension (typically C=256), and (H_c, W_c) is the spatial resolution of the descriptor grid. Let $\mathcal{B}$ be the set of valid batch indices used in this loss (i.e., those with sufficient verified correspondences), and $|\mathcal{B}|$ be its cardinality.

For each valid batch element $b \in \mathcal{B}$, we obtain N_b verified correspondences indexed by $i \in \{1, \ldots, N_b\}$. Following SuperPoint [11], each correspondence provides a pair of sampling locations $\{\mathbf{g}_{b,i}^0, \mathbf{g}_{b,i}^1\}$ from which we bilinearly sample and ℓ_2-normalize local descriptors $\mathbf{d}_{b,i}^0, \mathbf{d}_{b,i}^1 \in \mathbb{R}^C$.

To construct the contrastive objective, we form a cosine-similarity matrix $\mathbf{M}^{(b)} \in \mathbb{R}^{N_b \times N_b}$:

$$\mathbf{M}_{i,j}^{(b)} = \frac{\left(\mathbf{d}_{b,i}^0\right)^\top \mathbf{d}_{b,j}^1}{\tau_m}, \quad i, j \in \{1, \ldots, N_b\}, \tag{8}$$

where $\tau_m > 0$ is the temperature coefficient. The diagonal entries (i, i) correspond to the verified positive pair, while all off-diagonal entries act as in-batch negatives. Let the identity target vector be $\mathbf{y}^{(b)} \in \{1, \ldots, N_b\}^{N_b}$ with $\mathbf{y}_i^{(b)} = i$. We define a bidirectional loss by applying cross-entropy row-wise on $\mathbf{M}^{(b)}$ and on its transpose:

$$\mathcal{L}_{\text{desc}}^{(b)} = \frac{1}{2}\Big(\mathrm{CE}\big(\mathbf{M}^{(b)}, \mathbf{y}^{(b)}\big) + \mathrm{CE}\big((\mathbf{M}^{(b)})^\top, \mathbf{y}^{(b)}\big)\Big), \tag{9}$$

where $\mathrm{CE}(\cdot)$ denotes the standard cross-entropy loss with a softmax taken along each row of the input matrix. Finally, we average over all valid batch elements to obtain the overall descriptor objective:

$$\mathcal{L}_{\text{desc}} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \mathcal{L}_{\text{desc}}^{(b)}. \tag{10}$$

IV. Experiment

A. Data Curation

We construct COCO-Water from MS-COCO [20] training split by applying our physical formation in Eqs. 1– 5. Three difficulty levels are generated with varying attenuation coefficients $\beta \in \{0.08, 0.15, 0.28\}$ and a fixed first-order radial distortion coefficient $k_1 = -0.3$.

For the wavelength-dependent absorption in Eq. 2, we use fixed channel-wise scalings $[\tau_R, \tau_G, \tau_B]$. The choice of τ_R is based on the fact that red light is absorbed more rapidly in water compared to green or blue light. Accordingly, we reduce τ_R more aggressively to simulate realistic attenuation effects [28]. Specifically, the easy setting uses $[0.70, 0.95, 1.10]$, the medium setting uses $[0.60, 0.90, 1.10]$ and the difficult setting uses $[0.40, 0.80, 1.00]$.

For the backscatter term in Eq. (4), we adopt two fixed ambient light vectors $[B_R, B_G, B_B]$ (normalized to $[0, 1]$):

$$\mathbf{B}^{\text{light}} = \left[\frac{25}{255}, \frac{65}{255}, \frac{120}{255}\right], \quad \mathbf{B}^{\text{teal}} = \left[\frac{20}{255}, \frac{60}{255}, \frac{110}{255}\right].$$

We set $\mathbf{B} = \mathbf{B}^{\text{light}}$ for the easy configuration and $\mathbf{B} = \mathbf{B}^{\text{teal}}$ for both medium and difficult.

For the pseudo-depth range $(d_{\min}, d_{\max})$, we adjust it to control the effective optical path length. The depth range reflects the typical underwater visibility, where shallower depths increase visibility and deeper depths reduce it [7]. We set $(d_{\min}, d_{\max}) \in \{(1,5), (2,8), (3,12)\}$ meters for easy, medium, difficult, respectively. To avoid overly deterministic gradients, we mildly randomize the upper bound as $d_{\max} \leftarrow d_{\max} \cdot u, u \sim \mathcal{U}(0.8, 1.2)$.

We apply the same degradation operators (Eqs. (1)–(5)) and use the same parameter settings for OceanSim data.

B. Training Details

We perform progressive training with a two-step simulation adaptation followed by real-data refinement. We first train on COCO-Water and OceanSim data for 30 epochs (batch size 32) using Adam (lr=1e-4), with learning rate decay of 0.1 at epochs 15 and 25.

We then refine the descriptor head on the combined AQUALOC [3] and FLSea [29] for 10 epochs using Adam (lr=1e-4) with gradient clipping at 1.0 and a learning rate decay of 0.5 at epochs 4 and 7. To avoid sample leakage across splits, we adopt a sequence-level partition over AQUALOC and FLSea. The validation set consists of FLSea sequences *tiny canyon*, *coral table*, and *canyon2*, along with AQUALOC sequences *archaeology 09* and *harbor 06*. The test set consists of FLSea sequences *u canyon*, *small loop*, and *rock garden2*, together with AQUALOC sequences *archaeology 10* and *harbor 07*. All remaining sequences are used for training. GCDA descriptor refinement is an offline stage that takes approximately 50 hours on a single NVIDIA RTX 4090 GPU.

C. Quantitative Results

To rigorously assess keypoint matching performance, we establish a unified evaluation methodology that is consistently applied across all evaluated methods. Regarding the matching back-end, ORB and SIFT use mutual nearest-neighbor (NN) matching, whereas SuperPoint and AquaPoint (ours) use LightGlue. Our quantitative evaluation leverages five standard metrics: matches, inliers, ratio ($\sum$ inliers/ $\sum$ matches), valid rate (the proportion of image pairs yielding at least 8 verified inliers), and spatial distribution ratio ρ. The metric $\rho \in [0,1]$ measures how widely inliers cover the image plane. Specifically, we uniformly partition each image into a 12×12 grid and compute the fraction of grid cells that contain at least one inlier.

As summarized in Table I, AquaPoint significantly outperforms the baselines across both datasets. On AQUALOC, our method yields the highest number of putative matches (703.5) and verified inliers (587.7), substantially exceeding ORB (408.5 inliers), SuperPoint (332.9), and SIFT (273.6). Compared to SuperPoint under identical keypoint budgets and matching configurations, this corresponds to a 76.5% increase in verified inliers on AQUALOC (587.7 vs. 332.9).

TABLE I: **Quantitative comparison of feature matching performance.** AQL: AQUALOC, FLS: FLSea, SP: SuperPoint. All methods extract up to 1000 keypoints per frame. ρ denotes the spatial distribution ratio. t_0 denotes feature extraction time per image and t_1 denotes the total extraction and matching time per image pair.

Data	Method	matches ↑	inliers ↑	ratio ↑	valid rate ↑	ρ ↑	t_0/t_1 ↓
AQL	ORB	474.0	408.5	0.862	1.000	0.355	3.8 / **11.5**
	SIFT	307.8	273.6	**0.889**	1.000	0.456	25.6 / 53.8
	SP	408.5	332.9	0.815	0.892	0.663	3.4 / 15.2
	Ours	**703.5**	**587.7**	0.835	**1.000**	**0.849**	3.4 / 16.7
FLS	ORB	298.5	151.1	0.506	0.982	0.263	6.0 / 16.7
	SIFT	369.3	251.4	0.681	0.983	0.449	27.7 / 60.6
	SP	83.1	69.6	**0.837**	0.436	0.410	3.3 / **13.7**
	Ours	**513.2**	**409.9**	0.799	**0.985**	**0.691**	**3.3** / 15.9

Furthermore, AquaPoint achieves the highest spatial distribution ratio ($\rho = 0.849$), indicating a more uniform distribution of verified geometric constraints across the image plane.

This performance gap is even more pronounced on the challenging FLSea. AquaPoint establishes a substantial margin with 409.9 inliers, markedly surpassing SIFT (251.4) and ORB (151.1). Compared to SuperPoint, this result represents a 488.9% increase in verified inliers on FLSea (409.9 vs. 69.6) under identical keypoint budgets and matching configurations. Notably, while SuperPoint reports a high inlier ratio, it suffers from a severe degradation in robustness (valid rate 0.436) and spatial distribution ($\rho = 0.410$). This fragility indicates frequent correspondence collapse under severe hydro-optical distortions, where the detector fails to trigger sufficient responses in degraded regions. In contrast, AquaPoint consistently maintains a robust valid rate (0.985) and dense spatial coverage, providing reliable front-end associations critical for stable downstream SLAM performance.

In terms of efficiency, Table I indicates that the improved matching quantity and quality of AquaPoint do not come at the cost of prohibitive runtime. Its feature extraction time is essentially on par with ORB, and the total extraction and matching time remains comparable across datasets.

Fig. 3 provides qualitative evidence for the above. In addition to the matching visualization, we show an inlier-count heatmap, where each grid cell represents the number of inliers in that cell. While baseline methods such as ORB and SuperPoint exhibit highly clustered or sparse feature distributions, AquaPoint produces correspondences that are significantly denser and strictly multi-view geometrically verified, spanning the majority of the image grid.

Fig. 4 demonstrates the scalability of different algorithms under varying keypoint budgets. AquaPoint consistently yields a substantially higher absolute number of verified inliers as the Top-K threshold increases, while maintaining a highly stable inlier ratio.

D. Downstream SLAM Integration and Evaluation

1) Evaluation Setup and Datasets: To rigorously evaluate the practical impact of AquaPoint without being confounded

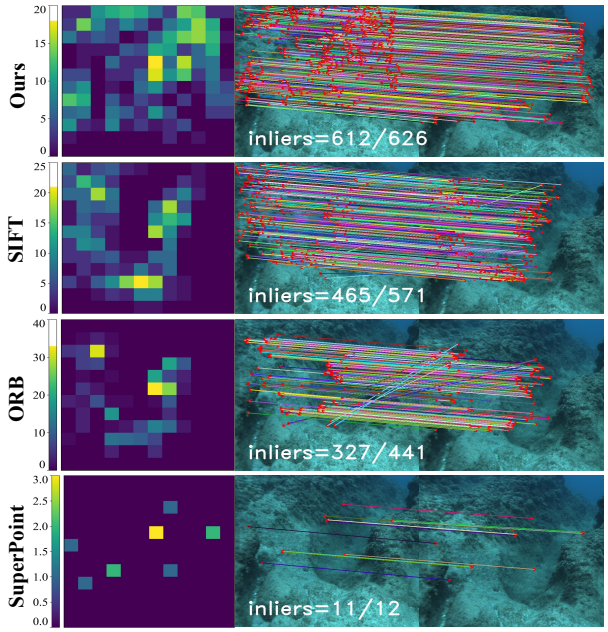
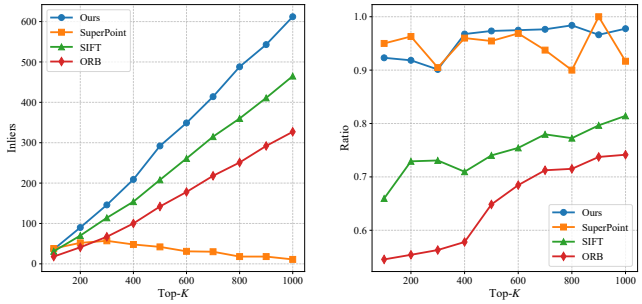


Fig. 3: **Representative matching visualizations and inlier-count heatmaps.** Heatmaps visualize the number of inliers per image patch, where the original image is partitioned into a 12×12 spatial grid. Our method yields denser and spatially uniform inliers with negligible mismatches.



(a) Number of Inliers (b) Inlier Ratio

Fig. 4: **Performance comparison of different algorithms under varying Top-K keypoint budgets. (a)** Absolute number of verified inliers. **(b)** Inlier ratio.

by the specific track-management sensitivities of sliding-window back-ends, we integrate AquaPoint into ORB-SLAM3 under a decoupled pure-odometry pipeline. By disabling global mapping components, we intentionally isolate the contribution of the front-end features to expose true tracking stability and drift. The neural keypoint budget is capped at $K = 1000$ per frame, employing LightGlue for robust feature association.

Evaluation is conducted on Tank dataset [30], which provides synchronized multi-sensor data and 6-DoF ground truth via an AprilTag-covered structure. We focus on the Structure-Easy (SE), Structure-Medium (SM), and Structure-Hard (SH), which maintain consistent ground-truth coverage while offering progressive levels of hydro-optical degradation and severe textureless observations.

SVIn2 [6], ORB-SLAM3 [4], and VINS-Fusion [5] (w. IMU) are evaluated as unmodified source systems. ORB-SLAM3 (w.o. IMU), SuperPoint (SP-VO), and AquaPoint are integrated into our decoupled pure-odometry configuration, while VINS (w.o. IMU) uses the original VINS-Fusion framework with the IMU deactivated.

2) Trajectory Analysis: We benchmark the trajectory quality on Tank dataset [30] across Structure sequences. We evaluate baseline methods under both visual-inertial (w. IMU) and pure vision (w.o. IMU) configurations. Following Table II, we report Absolute Trajectory Error (ATE), Absolute Rotation Error (ARE), and the trajectory completion ratio (TCR). All reported trajectory metrics are computed after aligning the estimated trajectory to the ground truth.

As demonstrated in Table II, while SP-VO achieves competitive accuracy in the well-lit SE sequence, it catastrophically fails in the more degraded SM and SH environments. This failure is fundamentally attributed to severe depth starvation and geometric filtering breakdown caused by scattering-induced noise.

ORB-SLAM3 (w.o. IMU) delivers acceptable performance in SE and SM, but completely loses tracking in SH sequence. Notably, VINS-Fusion (w.o. IMU) suffers from late initialization and trajectory fragmentation in degraded sequences, frequently diverging upon minor visual disturbances. Paradoxically, its ATE appears to decrease as environmental difficulty increases. We attribute this counterintuitive trend mainly to the joint effect of late initialization together with early termination. The reported error is computed only on a short surviving portion of the trajectory before catastrophic failure, where long-horizon drift has not yet accumulated. Therefore, TCR should be interpreted jointly with ATE and ARE to faithfully reflect robustness.

Interestingly, on Structure sequences, w.o. IMU systems outperform w. IMU systems in certain cases. This is not contradictory to the utility of inertial sensing, but rather reflects the dual role of IMU fusion. When visual constraints are sufficiently strong, pure-vision estimation can already be well-conditioned, whereas adding IMU introduces additional latent states (biases and their random walks) and extra calibration dependencies. Conversely, in the most challenging regimes where visual tracking intermittently collapses, the IMU prior can still be critical for bridging short-term dropouts and preventing immediate divergence.

In striking contrast, AquaPoint substantially fortifies tracking resilience against severe visual degradation, reducing ATE by 30.8% on average and boosting the trajectory completion ratio to 100% on challenging sequences, achieving state-of-the-art performance on Tank dataset Structure sequences. As qualitatively visualized in Fig. 5, AquaPoint's estimated trajectories closely align with the ground truth even under severe degradation, whereas baselines exhibit severe drift or catastrophic tracking failure.

It is worth noting that the Tank dataset also includes HalfTank and WholeTank sequences, where the vehicle traverses extended textureless regions and the 6-DoF GT is

TABLE II: **Trajectory evaluation on Tank dataset (Structure sequences).** "–" indicates catastrophic tracking failure or inability to initialize. "*" denotes results obtained from the Tank dataset paper [30]. **Units:** absolute trajectory error (ATE) in meters, absolute rotation error in degrees, and trajectory completion ratio (TCR) in %.

Sequence	Visual-Inertial System (w. IMU)						Pure Vision System (w.o. IMU)											
	SVIn2*		ORB-SLAM3*		VINS*		ORB-SLAM3			VINS			SP-VO			**AquaPoint (Ours)**		
	ATE	ARE	ATE	ARE	ATE	ARE	ATE	ARE	TCR	ATE	ARE	TCR	ATE	ARE	TCR	ATE	ARE	TCR
Structure-Easy	0.090	1.756	0.199	4.183	0.219	2.321	0.058	1.487	100%	1.891	83.139	100%	0.042	4.370	100%	**0.037**	**1.051**	**100%**
Structure-Medium	2.111	57.510	3.494	90.014	42359.413	65.957	0.059	2.010	100%	1.639	52.594	98.85%	–	–	0%	**0.044**	**1.381**	**100%**
Structure-Hard	3.589	23.439	2.933	102.615	5165.797	38.244	–	–	33.64%	1.289	35.356	86.30%	–	–	0%	**0.194**	**4.370**	**100%**

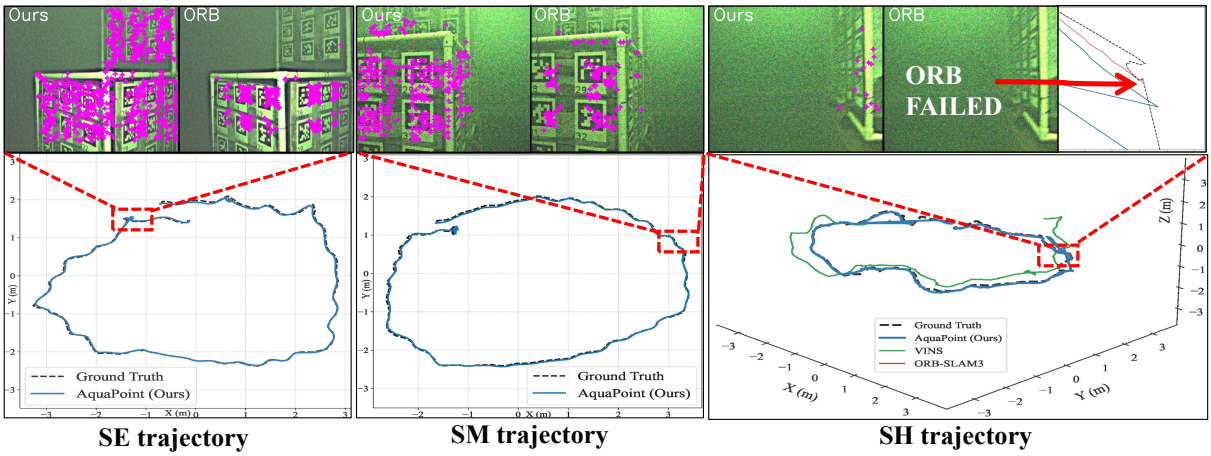


Fig. 5: **Trajectory comparison of AquaPoint and baselines against the ground truth on Structure sequences.** Notably, the results demonstrate AquaPoint's stable tracking, whereas ORB-SLAM3 suffers catastrophic tracking failure in the SH sequence.

only intermittently available when returning to the AprilTag structure. Since our configuration intentionally evaluates pure-vision drift under continuous geometric observability and consistent GT coverage, we report results on Structure sequences (SE/SM/SH), which retain persistent but highly degraded structural visibility and provide GT for most frames for quantitative benchmarking. Extending the evaluation to long, structureless segments with multi-sensor fusion (e.g., IMU/DVL) is left for future work.

E. Ablation Study

To isolate the contribution of each component in AquaPoint, we perform controlled ablations using the same evaluation configuration. Consistent with our main experiments, all variants employ LightGlue for feature association, with a maximum budget of $K = 1000$ keypoints.

1) OceanSim Fine-Tuning: We remove the OceanSim fine-tuning stage and directly perform real-data refinement after COCO-Water pretraining (w.o. OS FT). As reported in Table III, while performance on AQUALOC remains comparable to ours, robustness significantly deteriorates on the more challenging FLSea. Specifically, the valid rate drops, accompanied by a reduction in inliers, indicating that OceanSim provides useful structural priors for challenging underwater degradations.

2) GCDA Pseudo-Correspondence Mining: We replace the GCDA pseudo-label mining procedure with single-frame homography adaptation [11] without multi-view geometric verification (w. HA Supervision). In addition, we keep the GCDA mining pipeline but disable all verification checks during pseudo-correspondence filtering (w.o. Checks). As reported in Table III, w. HA Supervision causes a substantial degradation on FLSea, while w.o. Checks also consistently reduces robustness, indicating the necessity of GCDA's multi-view verified pseudo-correspondences and the effectiveness of the checks.

TABLE III: **Quantitative ablation study of the AquaPoint framework.** "w.o. OS FT" omits OceanSim fine-tuning; "w. HA Supervision" employs single-frame homography adaptation instead of GCDA pseudo-correspondence mining; "w.o. Checks" disables all verification checks in GCDA pseudo-label mining; "w. Hinge Loss" replaces InfoNCE with a margin-based hinge loss.

Data	Variant	matches ↑	inliers ↑	ratio ↑	valid rate ↑	ρ ↑
AQL	w.o. OS FT	**721.3**	**599.4**	0.831	1.000	0.830
	w. HA Supervision	680.0	570.7	**0.839**	1.000	0.843
	w.o. Checks	703.1	579.4	0.825	1.000	0.843
	w. Hinge Loss	693.9	580.5	0.837	1.000	0.848
	Ours	703.5	587.7	0.835	**1.000**	**0.849**
FLS	w.o. OS FT	502.5	400.1	0.796	0.942	**0.695**
	w. HA Supervision	425.6	345.8	**0.812**	0.946	0.635
	w.o. Checks	497.1	399.3	0.803	0.945	0.683
	w. Hinge Loss	486.2	391.4	0.805	0.986	0.667
	Ours	**513.2**	**409.9**	0.799	**0.986**	0.691

3) The InfoNCE Alignment Objective: We replace the InfoNCE objective with a margin-based hinge loss for descriptor learning (w. Hinge Loss). As reported in Table III, the AQUALOC performance remains highly comparable to ours. However, on FLSea, the number of inliers decreases, and the spatial distribution becomes less uniform, indicating that InfoNCE provides stronger discriminability and coverage, especially under more challenging underwater conditions.

V. Conclusion and Future Work

We presented AquaPoint, a self-supervised cross-domain framework for robust underwater local feature extraction. By synergizing the PGDB module and the GCDA module, AquaPoint mitigates photometric domain gaps, structural discrepancies, and ground truth scarcity. Evaluations on AQUALOC and FLSea, alongside integration into a pure-odometry ORB-SLAM3 system, demonstrate its superior matching accuracy, spatial observability, and trajectory tracking robustness under severe degradation compared to existing baselines. A limitation of our current framework is its reliance on air-calibrated distortion parameters to implicitly address depth-dependent water refraction. Future work will investigate distortion-aware training losses, joint detector-descriptor adaptation, and complete SLAM-level evaluation on diverse underwater sequences.

Acknowledgment

This work was supported in part by the 1+1+1 CUHK-CUHK(SZ)-GDSTC Joint Collaboration Fund, and in part by the NSFC under Grant No.62336005.

References

[1] S. Zhang, S. Zhao, D. An, J. Liu, H. Wang, Y. Feng, D. Li, and R. Zhao, "Visual SLAM for underwater vehicles: A survey," *Computer Science Review*, vol. 46, p. 100510, 2022.

[2] S. Xu, K. Zhang, and S. Wang, "AQUA-SLAM: Tightly Coupled Underwater Acoustic-Visual-Inertial SLAM With Sensor Calibration," *IEEE Transactions on Robotics*, vol. 41, pp. 2785–2803, 2025.

[3] M. Ferrera, V. Creuze, J. Moras, and P. Trouvé-Peloux, "AQUALOC: An underwater dataset for visual–inertial–pressure localization," *The International Journal of Robotics Research*, vol. 38, no. 14, pp. 1549–1559, 2019.

[4] C. Campos, R. Elvira, J. J. G. Rodriguez, J. M. M. Montiel, and J. D. Tardos, "ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual–Inertial, and Multimap SLAM," *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1874–1890, 2021.

[5] Y. Lin, F. Gao, T. Qin, W. Gao, T. Liu, W. Wu, Z. Yang, and S. Shen, "Autonomous aerial navigation using monocular visual-inertial fusion," *Journal of Field Robotics*, vol. 35, no. 1, pp. 23–51, 2018.

[6] S. Rahman, A. Q. Li, and I. Rekleitis, "Svin2: A multi-sensor fusion-based underwater slam system," *The International Journal of Robotics Research*, vol. 41, no. 11-12, pp. 1022–1042, 2022.

[7] D. Akkaynak and T. Treibitz, "Sea-Thru: A Method for Removing Water From Underwater Images," in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Long Beach, CA, USA: IEEE, 2019, pp. 1682–1691.

[8] M. She, F. Seegräber, D. Nakath, and K. Köser, "Refractive COLMAP: Refractive Structure-from-Motion Revisited," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024, pp. 12 816–12 823.

[9] D. G. Lowe, "Distinctive Image Features from Scale-Invariant Keypoints," *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.

[10] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in *2011 International Conference on Computer Vision*. Barcelona, Spain: IEEE, 2011, pp. 2564–2571.

[11] D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperPoint: Self-Supervised Interest Point Detection and Description," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. Salt Lake City, UT, USA: IEEE, 2018, pp. 337–33 712.

[12] J. Li, K. A. Skinner, R. M. Eustice, and M. Johnson-Roberson, "WaterGAN: Unsupervised Generative Network to Enable Real-time Color Correction of Monocular Underwater Images," 2017, arXiv:1702.07392 [cs].

[13] C. Li, C. Guo, W. Ren, R. Cong, J. Hou, S. Kwong, and D. Tao, "An underwater image enhancement benchmark dataset and beyond," *IEEE Transactions on Image Processing*, vol. 29, pp. 4376–4389, 2020.

[14] J. Song, H. Ma, O. Bagoren, A. Sethuraman, Y. Zhang, and K. A. Skinner, "OceanSim: A GPU-Accelerated Underwater Robot Perception Simulation Framework," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Hangzhou, China: IEEE, 2025, pp. 1526–1533.

[15] J. Revaud, C. De Souza, M. Humenberger, and P. Weinzaepfel, "R2d2: Reliable and repeatable detector and descriptor," *Advances in neural information processing systems*, vol. 32, 2019.

[16] W. Zhang, L. Dong, X. Pan, P. Zou, L. Qin, and W. Xu, "A Survey of Restoration and Enhancement for Underwater Images," *IEEE Access*, vol. 7, pp. 182 259–182 279, 2019.

[17] B. McGlamery, "A computer model for underwater camera systems," in *Ocean optics VI*, vol. 208. SPIE, 1980, pp. 221–231.

[18] D. Fuoli, Z. Huang, D. P. Paudel, L. Van Gool, and R. Timofte, "An Efficient Recurrent Adversarial Framework for Unsupervised Real-Time Video Enhancement," *International Journal of Computer Vision*, vol. 131, no. 4, pp. 1042–1059, 2023.

[19] F. Hidalgo and T. Bräunl, "Evaluation of Several Feature Detectors/Extractors on Underwater Images towards vSLAM," *Sensors*, vol. 20, no. 15, p. 4343, 2020.

[20] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common Objects in Context," in *Computer Vision – ECCV 2014*. Springer International Publishing, 2014, vol. 8693, pp. 740–755.

[21] J. S. Jaffe, "Computer modeling and the design of optimal underwater imaging systems," *IEEE journal of oceanic engineering*, vol. 15, no. 2, pp. 101–111, 2002.

[22] R. M. Haralick, S. R. Sternberg, and X. Zhuang, "Image analysis using mathematical morphology," *IEEE transactions on pattern analysis and machine intelligence*, no. 4, pp. 532–550, 1987.

[23] A. M. Reza, "Realization of the Contrast Limited Adaptive Histogram Equalization (CLAHE) for Real-Time Image Enhancement," *Journal of VLSI signal processing systems for signal, image and video technology*, vol. 38, no. 1, pp. 35–44, 2004.

[24] J. Shi *et al.*, "Good features to track," in *1994 Proceedings of IEEE conference on computer vision and pattern recognition*. IEEE, 1994, pp. 593–600.

[25] Z. Teed and J. Deng, "Raft: Recurrent all-pairs field transforms for optical flow," in *European conference on computer vision*. Springer, 2020, pp. 402–419.

[26] M. Ferrera, J. Moras, P. Trouvé-Peloux, and V. Creuze, "Real-time monocular visual odometry for turbid and dynamic underwater environments," *Sensors*, vol. 19, no. 3, p. 687, 2019.

[27] A. v. d. Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," *arXiv preprint arXiv:1807.03748*, 2018.

[28] D. Akkaynak, T. Treibitz, T. Shlesinger, Y. Loya, R. Tamir, and D. Iluz, "What is the space of attenuation coefficients in underwater computer vision?" in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 4931–4940.

[29] Y. Randall, *FlSea: Underwater visual-inertial and stereo-vision forward-looking datasets*. University of Haifa (Israel), 2023.

[30] S. Xu, J. Scharff Willners, J. Roe, S. Katagiri, T. Luczynski, Y. Petillot, and S. Wang, "Tank dataset: An underwater multi-sensor dataset for SLAM evaluation," *The International Journal of Robotics Research*, p. 02783649251364904, 2025.